NET732

Week 3: Cryptography

Practical

1. Preparing the environment

I opened Ubuntu and checked whether OpenSSL was installed using:

openssl version

```
maso-prince@maso-prince-VMware-Virtual-Platform:~$ openssl version
OpenSSL 3.5.5 27 Jan 2026 (Library: OpenSSL 3.5.5 27 Jan 2026)
```

My system showed OpenSSL version 3.5.5, confirming that it was installed and ready to use.

2. Generating a hash

I created a text file called test.txt and generated a SHA-256 hash using:

echo "CyberOps Week 3 Cryptography" > test.txt

openssl dgst -sha256 test.txt

```
maso-prince@maso-prince-VMware-Virtual-Platform:~$ openssl version
OpenSSL 3.5.5 27 Jan 2026 (Library: OpenSSL 3.5.5 27 Jan 2026)
maso-prince@maso-prince-VMware-Virtual-Platform:~$ mkdir -p ~/CyberOps/Week3
maso-prince@maso-prince-VMware-Virtual-Platform:~$ cd ~/CyberOps/Week3
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week3$ echo "CyberOps Week 3 Cryptography" > test.txt
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week3$ cat test.txt
CyberOps Week 3 Cryptography
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week3$
```

I observed that OpenSSL generated a unique hash for my file.

3. Verifying file integrity

I modified my text file and generated another hash:

echo "This file has been modified" >> test.txt

openssl dgst -sha256 test.txt

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week3$ echo "This file has been modified" >> test.txt
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week3$ openssl dgst -sha256 test.txt
SHA2-256(test.txt)= ef0115a8e4c8871c885c7d1e806d39550178b8a9911fd8d1c27e13e65ecbe095
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week3$
```

I noticed that the hash changed after modifying the file. This showed me how hashing can be used to detect changes and verify file integrity.

4. Creating a digital certificate

I used OpenSSL to generate a private key and a self-signed certificate:

openssl req -x509 -newkey rsa:2048 -keyout private.key -out certificate.crt -days 365 -nodes

I entered the required certificate details and successfully generated the files.

I then viewed my certificate using:

openssl x509 -in certificate.crt -text -noout

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week3$ openssl x509 -in certificate.crt -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            4f:90:04:0a:5c:59:e1:e2:0b:37:c7:95:c3:37:45:7a:30:27:ba:4e
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=ZA, ST=Gauteng, L=Johannesburg, O=CyberOps, OU=cyberops.local
        Validity
            Not Before: Oct 30 14:12:07 2026 GMT
            Not After : Oct 30 14:12:07 2027 GMT
        Subject: C=ZA, ST=Gauteng, L=Johannesburg, O=CyberOps, OU=cyberops.local
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (2048 bit)
                Modulus:
                    00:bf:c4:ba:78:b2:20:c3:2d:42:c2:95:c3:32:06:
                    47:f2:57:ca:3b:83:58:c9:89:e6:ea:7a:3f:c5:67:
                    0a:eb:da:b7:51:13:fe:57:a7:d5:77:db:e0:cb:e3:
```

I observed information such as the issuer, validity period and public key.

5. Saving evidence and logs

I created a Week 3 folder to store my practical files and evidence. I also created a commands.log file to record my practical activities.

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week3$ touch commands.log
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week3$ ls -l
total 12
-rw-rw-r-- 1 maso-prince maso-prince 1322 Oct 30 16:19 certificate.crt
-rw-rw-r-- 1 maso-prince maso-prince    0 Oct 30 16:27 commands.log
-rw------- 1 maso-prince maso-prince 1704 Oct 30 16:18 private.key
-rw-rw-r-- 1 maso-prince maso-prince   57 Oct 30 16:17 test.txt
```

My files included:

- test.txt

- private.key

- certificate.crt

- commands.log

6. Analysis and conclusion

During this practical, I learned how to use OpenSSL to generate hashes, verify file integrity and create digital certificates. I observed that changing a file changes its hash.

These techniques are important in SOC operations because security analysts use hashes to identify modified files and certificates to support secure communication.

I successfully completed the cryptography practical and saved my evidence. My next step is to practise digital signatures and certificate verification to improve my understanding of cybersecurity.